AIRSTREET · DESIGN DECISIONS REGISTER

AirStreet design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-24	Amish Chadha	Kit 1.9.1 upgrade: open decisions 1 to 5 (concept selection, diffusion tube cross-check, flagged hot hours, the 2030 EU limit, PR-017 and PR-018), proposed, awaiting Amish; STEM check fixes F1 to F5 recorded as decided under Amish's standing instruction of 2026-10-23 (AST-DDR-005)
0.4	2026-10-02	Amish Chadha	Value engineering restated after the A1 lightening (USD 302.50, USD 17.50 over); actions that carry the 2026-10-02 decisions outside the design listed
0.3	2026-10-02	Amish Chadha	Amish approved the recommendations for open decisions 1 to 8 (2026-10-02); moved to decisions made (AST-DDR-003 accepted)
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened with the open decisions from REVIEW.md, the

decision records and the build plan work

AirStreet design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

Open decisions

All proposed, awaiting Amish (kit 1.9.1 upgrade, 2026-10-24). Nothing in the design changes until he decides.

#	To decide	Options	Recommendation	What it affects in the build	Source
1	Concept selection	A, the current design (221 of 260); C, diffusion tubes with a battery PM logger (198); B, aspirated heated-inlet node (188)	A, kept as the design of record (no option within 10 %)	Nothing if A; a new design if B or C	AST-DDR-004, AST-OPT-001
2	Graft the runner-up's best idea: triplicate NO2 diffusion tubes at a few nodes as a check of the annual means and the street transfer bias	(a) yes, at the anchor node and one node in five; (b) yes, at every node; (c) no	(a): a printed hood and three tube clips on the cross arm, tubes changed every two to four weeks by the partner; laboratory cost about £8 per tube (2015 price)	Cross arm length and a hood part; a partner task	AST-OPT-001
3	Validity above the sensors' rated range (the NO2 cell is rated to 40 degC and 85 % RH; hot afternoons at a Dallas site and humid nights will be flagged, AST-STM-001 S33)	(a) accept flagged hours (the flags are made under the standing instruction, F2); (b) also seek a NO2 cell rated above 40 degC with the partner; (c) restrict PR-012 and the data claims to 40 degC	(a) now and (b) when the partner is agreed	None for (a); a supplier change for (b)	AST-DDR-005 F2, AST-STM-001 S15 and S33
4	PR-004 target: the recast Directive (EU) 2024/2881 lowers the NO2 annual limit to 20 ug/m3 from 2030	(a) keep 40 ug/m3; (b) classify against both 40 and 20 ug/m3; (c) 20 ug/m3 only	(b): the same uncertainty gives a band of 13.9 to 26.1 ug/m3 round 20 (at the directive's 20 degC)	Calculation and data products only	AST-REG-001 A2, AST-CAL-001 C
5	Text of PR-017 (open and garage buildable) and PR-018 (installation and service safety), added so that UR-014 and UR-015 each have a product requirement	(a) accept as written; (b) amend	(a)	None; both are met by design	AST-PRS-001

To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The band roll and housings close round both an 80 mm and a 200 mm pole with the saddle, and the band maker's torque for about 1,000 N of preload	The slip and twist checks assume 1,000 N per band	AST-CAL-001 [F7], [G2]
2	The NO2 front end board's size and mounting holes (model: 86 x 55 mm, holes 76 x 43 mm apart) and how the B4 sensor connects to it	They set the back wall bosses and the lead length to the sensor	AST-DDR-003, P8
3	The particle sensor's interface select pin and logic levels on its datasheet	Sets the terminal block wiring	Build plan section 3.10
4	Heat-set insert sizes (M5 and M3) and the hole size their maker gives	The printed holes are drawn 6.8 and 4.0 mm	AST-DWG-102, 106, 109
5	The M8 4-pin socket's panel hole and the M16 glands' hole size	The pod's printed holes are 8.2 and 16.2 mm	AST-DWG-106
6	FieldNode's enclosure lug kit positions match FieldNode's back plate pattern	The adapter plates copy that pattern	FND-DDR-003
7	The FieldNode panel's frame has a flat back lip at least 12 mm wide	FieldNode's panel clips bolt through it	FND-DDR-003
8	The pull-out strength of the M5 heat-set inserts in printed ASA (at least 200 N assumed in the STEM check)	The pod hangs on two of them	AST-STM-001 S26
9	The insect mesh aperture against the smallest local insects	Insects and spiders in the pod	AST-STM-001 S31

Value engineering

Value-engineering target: USD 285. Estimated cost of the constructable design: USD 302.50 (USD 17.50 over the target).

The target is `budget_usd`, a hypothetical control target, not a limit, and covers the sensor head only; with the FieldNode core, which is costed in FieldNode, the node is USD 433.50. Main cost drivers and savings worth trying:

- The rise from USD 285 came from making the design constructable (AST-DDR-003): line 3, band stock, adapter plates and cross arm (USD 4.50, after the A1 lightening: the thinner rail saves about USD 0.50 and the saddles are repriced at about USD 3.00 of filament at 40 % infill); line 9, probe tube (USD 1); line 10, probe lead with its M8 plug (USD 4); and line 11, inserts, glands, probe socket, terminal block and fixings (USD 8). Line 8 is back to USD 9.00: the 1.5 mm shield plates save about as much filament as the spacers use.
- Savings worth trying: the open two-channel NO2 front end (about USD 35 cheaper, PCB work that is on hold with TRL 4) would recover the USD 17.50, and cheaper band hardware and fixings can be sought at purchase.

Actions that carry the decisions outside the design

These follow from decisions already made; none is an open decision.

#	Action	Who	From
1	Ask the pole owner of the first site to confirm a load of 4.0 kg on the pole (3.78 kg with the sun shield as designed) and about 125 N of wind load 3.3 m up	Amish, once a site is chosen	Mass decision (R13), 2026-10-02
2	Approach the first candidate partner, the Texas Commission on Environmental Quality's Dallas-Fort Worth monitoring network, with a local university or city partner; nothing is agreed yet	Amish	First partner decision, 2026-10-02
3	When the partner is agreed, fix the radio band and FieldNode antenna in the BOM; for US915, restate R10 against the 400 ms dwell limit (SF9 or lower)	Next design update after the partner is agreed	Radio band decision, 2026-10-02
4	Send FieldNode the adapter plate request and track its confirmation of port B's always-on 5 V and the 5 min private-gateway interval	Cross-repo (see REVIEW.md)	Decisions of 2026-10-02 on FieldNode
5	Propose the common M12 pin assignment to FieldNode, with CurbCount; once FieldNode agrees, update the pod wiring in build plan section 3.10 and its wiring picture	Cross-repo (see REVIEW.md)	Pin assignment decision, 2026-10-02
6	Check LoRaWAN signal strength with the pod fitted; move the pod 8 mm left in the model and pictures only if the check shows a loss	TRL 4 (on hold)	Antenna whip decision, 2026-10-02

Decisions made

Date	Decision	Decided by	Record
2026-10-24	F1: NO2 noise per sample taken from the NO2-B43F data sheet (7.5 ppb, 1 sigma), averaged over the 12 independent records of an hour, with a 5 % slope term (the last two after the independent review); hourly total 5.45 ppb, MAE about 4.3 ppb	Decided under Amish's standing instruction of 2026-10-23 ("propose and implement the best fix ... and record it in DEC as decided under that instruction"); reversible by Amish	AST-DDR-005 F1
2026-10-24	F2: rated-range flags in every record (shield air above 35 degC, RH above 80 % for PM or 85 % for NO2 and below 20 % or 15 %, below 10 degC for PM); flagged hours published with the flag	Decided under the standing instruction of 2026-10-23	AST-DDR-005 F2
2026-10-24	F3: the NO2 cell is replaced every 12 months at the pod exchange, or at 18 or 24 months where the site's ozone record keeps its filter under 500 ppm h	Decided under the standing instruction of 2026-10-23	AST-DDR-005 F3
2026-10-24	F4: band drawn 12.7 x 0.6 mm and read by the calculation	Decided under the standing instruction of 2026-10-23	AST-DDR-005 F4
2026-10-24	F5: D3 model fit details (pod boss webs, M3 insert holes in the back wall bosses, hollow shield spacers, probe lead route, M5 x 14 arm screws with washers)	Made under Amish's 2026-09-30 instruction to make the design physically buildable	AST-DDR-005 F5
2026-09-25	TRL 2 review items D1 to D9: budget covers the sensor head only; pitch wording; NO2 collocation plan; NO2-B43F class sensor; ISB class front end bought first; SPS30 class particle sensor; 3.0 m inlets; 5 min records with raw signals; open data	Amish: "i accept all your recommendations, go with them across all repos."	AST-DDR-001, AST-DDR-002
2026-09-25	`budget_usd` $285 for the sensor head; FieldNode back plate left off with the enclosure and bracket on the AirStreet rail; frontal area limit 0.15 m²; R3 (WHO guideline) withdrawn and kept as research; 60 s particle run; 5 min records on a private gateway and 15 min on The Things Network; FieldNode sun shield at sites above 45 °C; pre-collocated pod exchange for service	Amish, same instruction: go with recommendation	AST-DDR-002, N1 to N7
2026-10-02	Design for construction accepted as a whole: the changes P1 to P10 (adapter plates, 140° V-saddles, bands in saddle grooves, cross arm, two-part pod, cable entries, shield spacers and probe tube) and their knock-on changes, as made	Amish: "i approve your recommendations for all 555 open decisions."	AST-DDR-003, Table 1
2026-10-02	Mass (R13): option (c). The R13 limit is set at 4.0 kg including FieldNode's sun shield, the pole owner is asked to confirm that load, and the four lightening steps (saddles at 40 % infill, 40 x 4 mm rail, 1.5 mm shield plates, windows in the adapter plates) are carried into the TRL 4 build to give margin	Amish: "i approve your recommendations for all 555 open decisions."	AST-DDR-003, A1
2026-10-02	Antenna whip: accept the 15 mm clearance to the pod's drip lid and check signal strength with the pod fitted at	Amish: "i approve your recommendations for all 555 open decisions."	AST-DDR-003, A3

Date	Decision	Decided by	Record
	TRL 4; move the pod 8 mm left only if that check shows a loss		
2026-10-02	First partner chosen by one rule: a city or air agency that runs a regulatory NO2 reference station at a street-level site AirStreet can mount beside, and that will share its data. First candidate to approach: the Texas Commission on Environmental Quality's Dallas-Fort Worth monitoring network, with a local university or city partner	Amish: "i approve your recommendations for all 555 open decisions."	AST-DDR-001, O1
2026-10-02	Agree with FieldNode: send FieldNode the request to mount its enclosure, plate clips and sun shield on AirStreet's two adapter plates, and to keep its model stable since AirStreet carries a copy	Amish: "i approve your recommendations for all 555 open decisions."	AST-DDR-002 N2, AST-DDR-003
2026-10-02	Port B's always-on 5 V supply and the 5 min interval on a private gateway: closed as already decided on 2026-09-25 (AST-DDR-002, N5 and its cross-repo actions); FieldNode's confirmation is tracked as a cross-repo action, not a decision	Amish: "i approve your recommendations for all 555 open decisions."	AST-DDR-002
2026-10-02	FieldNode port pin assignment: propose to FieldNode one common assignment for both M12 ports (supply, ground, I2C data, I2C clock and one spare line), with each port's supply voltage set per project by its supply module (5 V on both ports for AirStreet)	Amish: "i approve your recommendations for all 555 open decisions."	FND-DDR-001, O2
2026-10-02	Radio band set with the partner of the first-partner decision above (not with the adapter plates, as the register said): US915 for a North American partner, EU868 for a European one, with the matching FieldNode antenna	Amish: "i approve your recommendations for all 555 open decisions."	FND-DDR-001, O1